

Trainings ▾

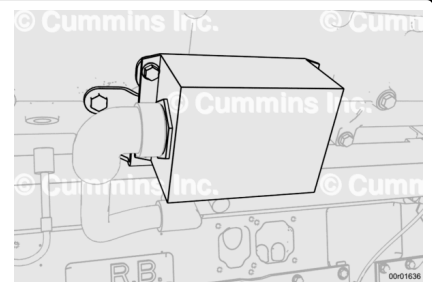
General Information

Thermocouple

QSK50

This engine has two exhaust gas temperature sensor converters: one on the right bank and one on the left bank.

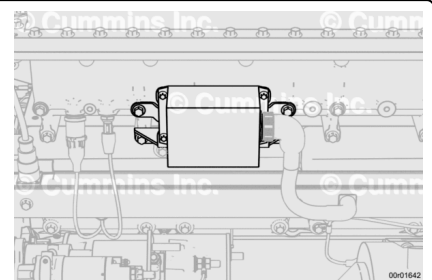
The exhaust gas temperature converter is mounted at the front of the intake manifolds.



QSK60

This engine has two exhaust gas temperature sensor converters: one on the right bank and one on the left bank.

The exhaust gas temperature converter is mounted at the front of the intake manifolds.



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Disconnect the battery cables. Refer to Procedure 013-009 in Section 13.
(/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
This procedue is found in QSK45 and QSK60, Service

Manual, Bulletin 4021530 or the K38, K50 and QSK50, Service Manual, Bulletin 4021528.

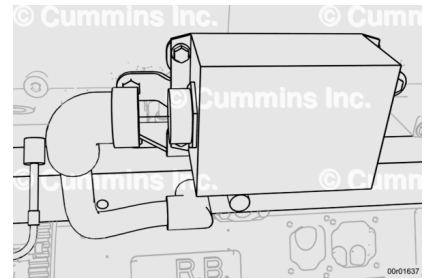
- Disconnect the engine harness cable from the exhaust gas temperature converter box.

Remove

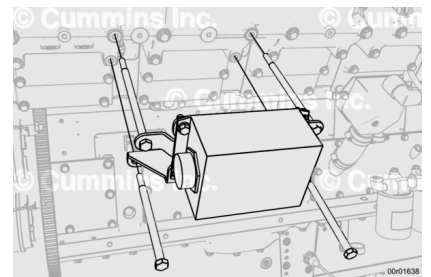
Thermocouple

QSK50

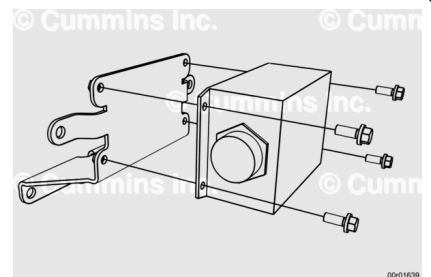
Disconnect the exhaust gas temperature sensor converter from the main wiring harness connector.



Remove the four outer mounting capscrews to remove the exhaust gas temperature converter and mounting bracket.

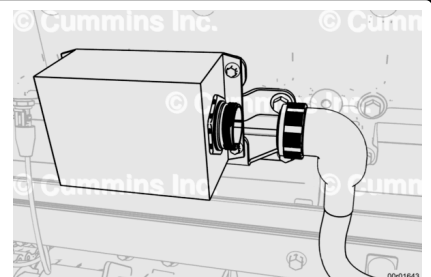


Remove the four mounting capscrews and the exhaust gas temperature converter from the mounting bracket.

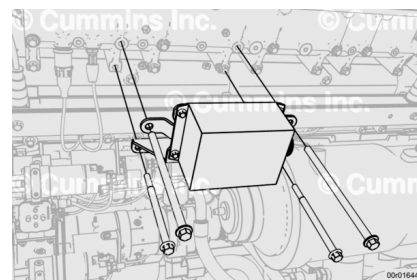


QSK60

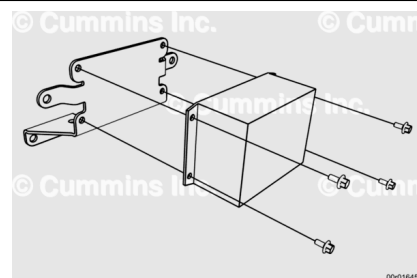
Disconnect the exhaust gas temperature sensor converter from the main wiring harness connector.



Remove the four outer mounting capscrews to remove the exhaust gas temperature converter and mounting bracket.



Remove the four mounting capscrews and the exhaust gas temperature converter from the mounting bracket.



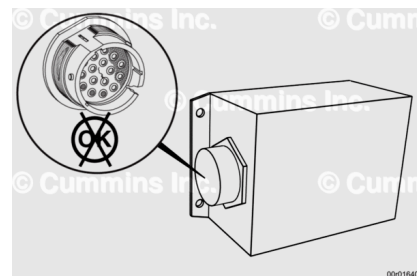
Clean and Inspect for Reuse

Thermocouple

QSK50 and QSK60

Inspect the converter and replace if:

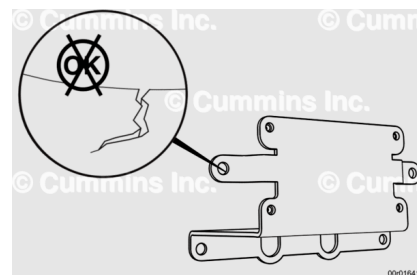
- Shell cracked or broken
- Seals missing or damaged
- Terminals contaminated with dirt, debris, or moisture
- Terminals corroded, bent, broken, pushed back, or expanded.



Inspect the mounting hardware.

Replace components if:

- Bent
- Cracked.



Install

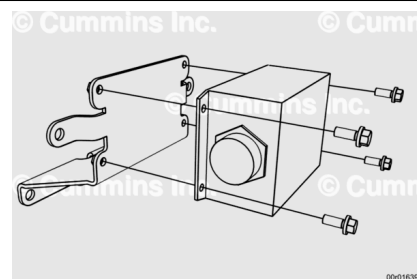
Thermocouple

QSK50

Install the exhaust gas temperature converter to the mounting bracket with four mounting capscrews.

Tighten the capscrews.

Torque Value: 10 n•m [89 in-lb]



Install the exhaust gas temperature converter and mounting bracket using the four mounting capscrews.

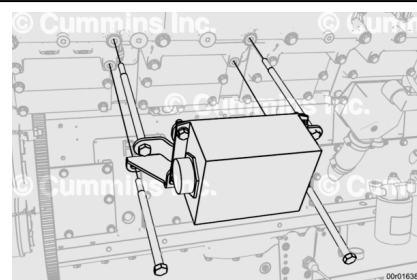
Tighten the capscrews.

Torque Value:

Aluminum 45 n•m [33 ft-lb]

Torque Value:

Cast Iron 65 n•m [48 ft-lb]

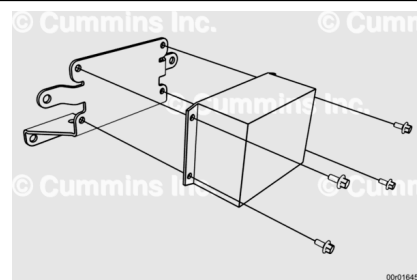


QSK60

Install the exhaust gas temperature converter to the mounting bracket with four mounting capscrews.

Tighten the capscrews.

Torque Value: 10 n•m [89 in-lb]



Install the exhaust gas temperature converter and mounting bracket using the four mounting capscrews.

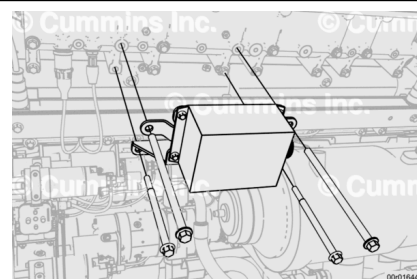
Tighten the capscrews.

Torque Value:

Aluminum 45 n•m [33 ft-lb]

Torque Value:

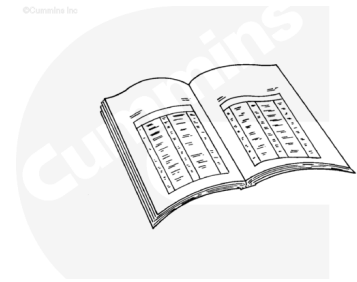
Cast Iron 65 n•m [48 ft-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



ck800wa

- Connect the engine harness cable to the exhaust gas temperature converter box.
- Connect the battery cables. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>) This procedure is found in QSK45 and QSK60, Service Manual, Bulletin 4021530 or the K38, K50 and QSK50, Service Manual, Bulletin 4021528.
- Connect recommended Cummins® electronic service tool or equivalent.
- Operate the engine and check for active fault codes.